Bt605 past papers for final term

Subject: Bt605
(past papers)

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1)what are Biosecurity plan 3 ?

Ans: Biosecurity plan: •written plan-prevent the introduction and spread of disease to farm •daily operation procedures •disinfecting procedures-part of the plan. – Responsibilities: • principle investigator plan implementation-workers following the plan-training • lab workers • responsible official • campus security staff • management services RO: • contacted if biological agent is theft or lost • contacted agencies if there is threat or spill • training Other aspects: • risk assessment • physical protection • personnel protection • pathogen accountability • emergency response.

2)Biohazard is classified into four elements name them write their categories and name their substance 5 ?

Ans:

The United States Centers for Disease Control and Prevention (CDC) categorizes various diseases in levels of biohazard, Level 1 being minimum risk and Level 4 being extreme risk. Laboratories and other facilities are categorized as BSL (Biosafety Level) 1-4 or as *P1* through *P4*for short (Pathogen or Protection Level).

Biohazard Level 1

Biohazard Level 2

Biohazard Level 3

Biohazard Level 4

- **Biohazard Level 1**: Bacteria and viruses including *Bacillus subtilis*, canine hepatitis, *Escherichia coli*, varicella (chicken pox), as well as some cell cultures and non-infectious bacteria. At this level precautions against the biohazardous materials in question are minimal, most likely involving gloves and some sort of facial protection.
- **Biohazard Level 2**: Bacteria and viruses that cause only mild disease to humans, or are difficult to contract via aerosol in a lab setting, such as hepatitis A, B, and C, some influenza A strains, Lyme

disease, salmonella, mumps, measles, scrapie, dengue fever, HIV. Routine diagnostic work with clinical specimens can be done safely at Biosafety Level 2, using Biosafety Level 2 practices and procedures. Research work (including co-cultivation, virus replication studies, or manipulations involving concentrated virus) can be done in a BSL-2 (P2) facility, using BSL-3 practices and procedures.

- **Biohazard Level 3**: Bacteria and viruses that can cause severe to fatal disease in humans, but for which vaccines or other treatments exist, such as anthrax, West Nile virus, Venezuelan equine encephalitis, SARS virus, MERS coronavirus, hantaviruses, tuberculosis, typhus, Rift Valley fever, Rocky Mountain spotted fever, yellow fever, and malaria.
- **Biohazard Level 4**: Viruses that cause severe to fatal disease in humans, and for which vaccines or other treatments are *not* available, such as Bolivian hemorrhagic fever, Marburg virus, Ebola virus, Lassa fever virus, Crimean–Congo hemorrhagic fever, and other hemorrhagicdiseases and rishibola. Variola virus (smallpox) is an agent that is worked with at BSL-4 despite the existence of a vaccine, as it has been eradicated. When dealing with biological hazards at this level the use of a positive pressure personnel suit, with a segregated air supply, is mandatory. The entrance and exit of a Level Four biolab will contain multiple showers, a vacuum room, an ultraviolet light room, autonomous detection system, and other safety precautions designed to destroy all traces of the biohazard. Multiple airlocks are employed and are electronically secured to prevent both doors opening at the same time. All air and water service going to and coming from a Biosafety Level 4 (P4) lab will undergo similar decontamination procedures to eliminate the possibility of an accidental release. Currently there are no bacteria classified at this level.

3)Write a note on Threats of biological weapons ?

Threats of biological weapons History: • 2001 - Tom Dachle –received a letter of anthrax • 23 members of the staff • 5 police officers • positive-nasal swabs • citizen-stockpiling - Ciprofloxacin Potential threats: • naturally present in the environment • no major infrastructure • no manpower • easier and faster

Potential threats: • cost-effective than poor's man atomic bomb • cover large area • difficult to diagnose and treat • high mortality and mobility Potential threats: • possibly contagious-small pox, plaque, viral hemorrhagic fever • create panic • weapon of mass disruption Potential threats: • detection devices, equipment for surveillanceexpensive • not present in many countrie

4)Write a efforts to mitigate biological threats?

Ans: Efforts to mitigate biological threats Bioweapons: • any infectious agent used intentionally to cause harm to others • planning of an effective biowardefence-difficult task • nation and scientific community Defence against bioweapons: • international

cooperation • transfer of technology • support national actions. BTWC: • BTWC-1972 • institutionally weak • implementation is ineffective • without investigation Biological threats: • terrorist, non-state actors • misuse of technologies • theft from lab • religious extremists • locally hired agents • frustrated cult • Efforts to mitigate: • BTWC Act • designation of focal point • central implementation authority • biosafety/biosecurity • code of conduct /awareness

Pakistan's approach: •front-line of "War on terror" •financial and human loss •bioweapons are not the part of security matrix Pakistan's approach: •tremendous progress •legislations •administrative measures NCGLs-2007: •national core group of life sciences •biosafety -syllabus •final review by HEC

PBSA-2008: •Pakistan biological safety association •train the trainers •seek for biological certification •risk assessment/risk management/ lab designing Efforts: •NIH-WHO •Export-control Act 2004-re-export, overexport •transshipment-goods, technologies, equip •14 yrs-imprisonment /Rs.5 million both

5) Name the things for protection in first aid kit Name lab safety things 2

Ans: First aid: Definition: •medical treatment--- time and place of an accident approved method treat the casualty before doctor's care for treatment Contains??: •first aid box •protective clothing •safety equipment •eye irrigation

First aid box: • made up of material-keep the content dust and damp free • white cross with green box • prominent position • easily recognizable Box contains??: • first aid manual • bandages • sterile dressings • safety pins Protective equipment: • mouth piece for mouth-to-mouth resuscitation • gloves-protections • clean up kit for blood spills

6)How immunization can be provided to fight against pathogens.?

Ans: Immunization of staff Introduction: • discuss with workers • vaccines • therapeutic drugs-after exposure.

The environment contains a wide variety of potentially harmful organisms (pathogens), such as bacteria, viruses, fungi, protozoa and multicellular parasites, which will cause disease if they enter the body and are allowed to multiply. The body protects itself through a various defence mechanisms to physically prevent pathogens from entering the body or to kill them if they do.

The immune system is an extremely important defence mechanism that can identify an invading organism and destroy it. Immunisation prevents disease by enabling the body to more rapidly respond to attack and enhancing the immune response to a particular organism.

7) what is cybernetics (2)

Ans:Cybernetics is a transdisciplinary[1] approach for exploring regulatory systems—their structures, constraints, and possibilities. Norbert Wiener defined cybernetics in 1948 as "the scientific study of control and communication in the animal and the machine."[2] In the 21st century, the term is often used in a rather loose way to imply "control of any system using technology." In other words, it is the scientific study of how humans, animals and machines control and communicate with each other.

Q9)applications of national biosafety rule?

Biosafety Application of National biosafety rules

Application: •manufacture, import and storage of microorganisms •gene technological products for research •field trial of GMOs •import, export, sale and purchase of GMOs

9) functions of technical adversary committee ?5

Ans: Functions: • examine applications and recommend to NBC • review and control of safety measures • review research methodologies • monitor release of GMOs/products into environment • provide information to NBC about approved projects • supervise the implementation of terms and conditions.

12)Biosafety precautions what you know about risk management and what safety measure required for tb lab. ?

Ans: Risk: • noise measurement surveys • specify areas • lab equipments(laser) • animal house Control: • barriers around noisy equipments • barrier between noisy areas and work areas • hearing conservation program • medical monitoring program

13)discus transport security element. write some activities of bio safety officer?

Ans:

Biosafety

. 100 Biosafety Biosafety officer Biosafety officer: • appointment • ensure biosafety programs and policies • small scale-technical staff • microbiology, biochemistry, basic biological sciences – Duties: • apply National or International rules, regulations/guidelines • assist lab in developing standard safety procedures • knowledge of lab, clinical practices

Duties: • knowledge of devices/engineering principles • maintenance facilities • communicate with administrative, technical and support personnel Activities: • biosafety consultation • biosecurity consultation • technical compliance consultation Biosafety

audits: • technical methods • procedures • protocols • biological agents • materials • equipments

Activities: • discussion-violation with appropriate persons • biosafety training • continue education in biosafety • investigation of accidents Activities: • decontamination of spills • proper waste management • decontamination of apparatus • awareness of community attitudes –Activities: •procedures for import/export of pathogens •review biosafety aspects of research work •institution of a system to deal with emergencies.

14) Eutrophication ?

Eutrophication (from Greek *eutrophos*, "well-nourished"),[1] or hypertrophication, is when a body of water becomes overly enriched with minerals and nutrients that induce excessive growth of plants and algae.[2] This process may result in oxygen depletion of the water body.[3]One example is the "bloom" or great increase of phytoplankton in a water body as a response to increased levels of nutrients. Eutrophication is almost always induced by the discharge of nitrate or phosphate-containing detergents, fertilizers, or sewage into an aquatic system.

15)-Biosafety & Bioethics?

Biosafety: • prevention of large-scale loss-biological integrity • ecology-human health • man made unicellular organisms-effect on biomass • enter into food chain.

Bioethics: • due to health and social benefits • to individual and families living free of mitochondrial disorders • parents having the preference to have genetically related children Bioethics: • novel tech prove to be safe • acceptable and effective as treatments • would be ethical for parents to use them.

16) 2. Functions of physical security?

ANS: Biosecurity Physical security Introduction: • protection of people/ hardware/programs • data / networks • physical events-terrorism/disasters .Physical security elements Elements: • obstacles placed in the way of attackers • surveillance and notification system • methods to recover quickly from disaster

Physical security elements Obstacles: • fencing • wall • multiple locks • fireproof safes • water sprinkles Surveillance/notification system: • heat detector • smoke detector • lighting • alarms • cameras Recovery: • repairment • hiring additional security • cameras

17) funcation of technical advisory committee?

Functions: • examine applications and recommend to NBC • review and control of safety measures • review research methodologies • monitor release of GMOs/products into environment • provide information to NBC about approved projects • supervise the implementation OF terms and condition.

18) What are Malicious risk groups, explain it? (10)

Biosecurity Characterize risk Malicious risk groups: • non-pathogenic • low malicious use risk (LMUR) • moderate malicious use risk (MMUR) • high malicious use risk (HMUR)/extreme malicious use risk (EMUR). Description: •non-pathogenic-inherent hazardous •no/insignificant consequences •LMUR - low consequences •most biological agents Example: •Mycobacterium leprae •gram positive rods/ non-spore formers •organism grow slowly-generation time 30 days •not highly virulent (LMUR)

Description: •MMUR-can't deploy as biological weapons •low / moderate consequences •low / moderate economic impact •many current agents evaluated as MMUR Example: •C.immitus – fungus •desert fever •cure without treatment/life-threatening cases •asymptomatic •biosafety level 3-MMUR

Description: • HMUR • national/international consequences • high casualties • high economic impact • Bacillus anthracis Description: • EMUR-HMUR • not found in the nature • high security measures • eradicated • genetically engineered agents Examples: • Variola major virus • small pox • highly virulent /contagious/stable in droplets • eradicated by vaccine • GM virus – more virulent.

19)3. Test tube baby? (5)

is one of the medical procedures which help people to deal with the infertility problems. Test tube baby is a term that refers to a child that is conceived outside the women's body by a scientific process known as In-Vitro fertilization or IVF treatment. This entire process is done in a laboratory. In this process the eggs are taken from the mother's ovary and fertilized by the sperms from the father.

The fertilised egg is cultured for 2–6 days and allowed to divided 2-4 times inside a test tube (hence the name test tube baby) These eggs are then returned back to the mother's uterus where it can be developed normally, this is done with the intention to establish a successful pregnancy.

Test tube baby procedure has greatly helped women having infertility problems that are untreatable to give birth to healthy babies.

20). What are serveilence elements?

4 Either malaria, tuberculosis or HIV surveillance, you must have a case definition prodromic, syndromic, sentinel, case-based,....community-based surveillance, genetic or genomic surveillance. Depending on your study it(surveillance) can be use as intervention or as a tool. thus it require 1) case definition, 2) early detection and investigation through epidemiological, lab, field and ecological/climatic data collection, 3) confirmation and reporting at different levels, 4) risk information communication and/or early warning alarm systems, 5) rapid response, which is something else on itself

and all these are done over time in a s sustained manner (time series: periodic, seasonal or continuous) as this require ample funding support.

This can be applied to any of the disease about depending on the phase where the indicators or performance metrics vary from prevention, preparedness, control or containment to elimination phases.

OR

public health surveillance has focused on communicable diseases. However, the scope of modern health surveillance extends to a broader range of health conditions such as injuries, occupational health and safety, environmental exposures, birth defects, addictions, chronic diseases, mental health and health behaviours. The list of organizations conducting surveillance has expanded as well. Although the traditional players remain - public health agencies, other health care organizations, and hospitals - in some countries in the Caribbean, surveillance is also conducted by academics, treatment centres and non-government organizations. The settings for surveillance have also expanded: recent examples include schools and shelters following natural or man-made disasters and displaced person's camps.

All surveillance systems must collect, review, analyse and transform data into useful information for public health action. However, some surveillance systems are more effective than others. What makes a system effective? There is a series of features or characteristics common to all effective surveillance systems (*Maxcy- Rosenau -Last, 1998*)

21) How risk can be reduced through risk management? 5

All risk management processes follow the same basic steps, although sometimes different jargon is used to describe these steps. Together these 5 risk management process steps combine to deliver a simple and effective risk management process.

Step 1: Identify the Risk. You and your team uncover, recognize and describe risks that might affect your project or its outcomes. There are a number of techniques you can use to find project risks. During this step you start to prepare your Project Risk Register.

Step 2: Analyze the risk. Once risks are identified you determine the likelihood and consequence of each risk. You develop an understanding of the nature of the risk and its potential to affect project goals and objectives. This information is also input to your Project Risk Register.

Step 3: Evaluate or Rank the Risk. You evaluate or rank the risk by determining the risk magnitude, which is the combination of likelihood and consequence. You make decisions about whether the risk is acceptable or whether it is serious enough to warrant treatment. These risk rankings are also added to your Project Risk Register.

Step 4: Treat the Risk. This is also referred to as Risk Response Planning. During this step you assess your highest ranked risks and set out a plan to treat or modify these risks to achieve acceptable risk levels. How can you minimize the probability of the negative risks as well as enhancing the opportunities? You create risk mitigation strategies, preventive plans and contingency plans in this step. And you add the risk treatment measures for the highest ranking or most serious risks to your Project Risk Register.

Step 5: Monitor and Review the risk. This is the step where you take your Project Risk Register and use it to monitor, track and review risks.

Risk is about uncertainty. If you put a framework around that uncertainty, then you effectively de-risk your project. And that means you can move much more confidently to achieve your project goals. By identifying and managing a comprehensive list of project risks, unpleasant surprises and barriers can be reduced and golden opportunities discovered. The risk management process also helps to resolve problems when they occur, because those problems have been envisaged, and plans to treat them have already been developed and agreed. You avoid impulsive reactions and going into "fire-fighting" mode to rectify problems that could have been anticipated. This makes for happier, less stressed project teams and stakeholders. The end result is that you minimize the impacts of project threats and capture the opportunities that occur.

If you enjoyed this article our Risk Management Online Course expands on these concepts and gives you the practical skills to impress your boss by developing a comprehensive risk management process.

22)define Biosecurity 3 ?

Ans) Biosecurity: • the emergence of biosecurity-critical policy area in 21st century • revolutionary changes-transformed • government approaches.

Biosecurity

US biosecurity regulations implementation: • new biosecurity legislation is a large body of work • success is critical to large number of clients/ stakeholders they understand the implementation and regulations.